

dotdraw

September 15, 2026

This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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```
[1]: import casadi as ca
     from casadi.tools import *
```

An SX graph

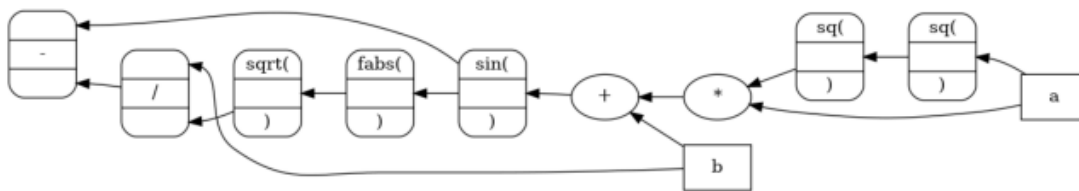
```
[2]: a = ca.SX.sym("a")
     b = ca.SX.sym("b")

[3]: c = ca.sin(a**5 + b)

[4]: c = c - b/ ca.sqrt(ca.fabs(c))
     print(c)
```

```
@1=sin(((a*sq(sq(a)))+b)), (@1-(b/sqrt(fabs(@1))))
```

```
[5]: dotdraw(c)
```

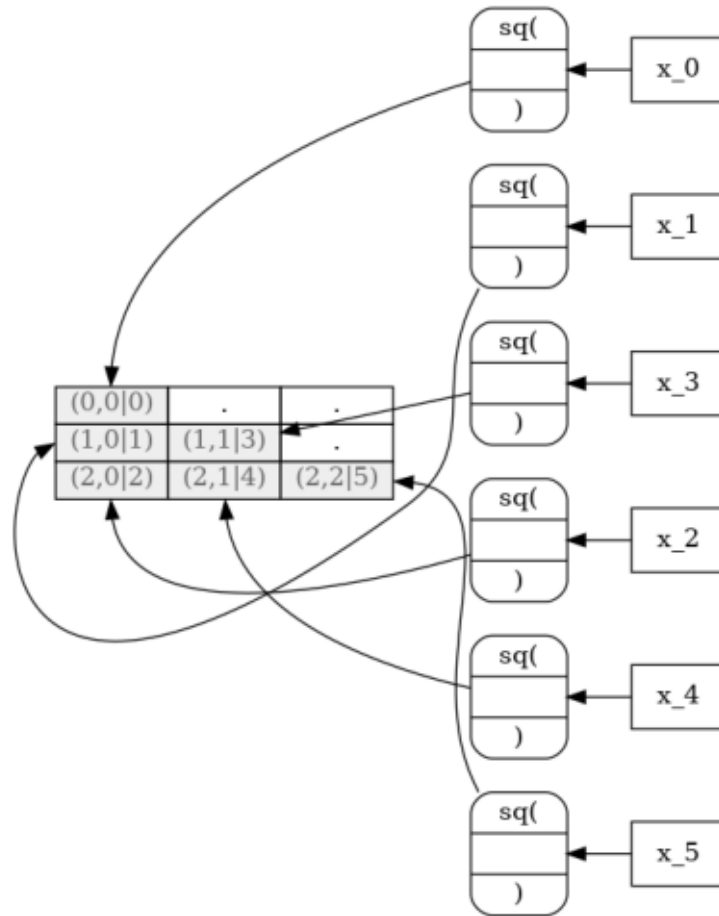


An SX

```
[6]: dotdraw(ca.SX.sym("x",ca.Sparsity.lower(3)))
```

x_0	.	.
x_1	x_3	.
x_2	x_4	x_5

```
[7]: dotdraw(ca.SX.sym("x",ca.Sparsity.lower(3))**2)
```



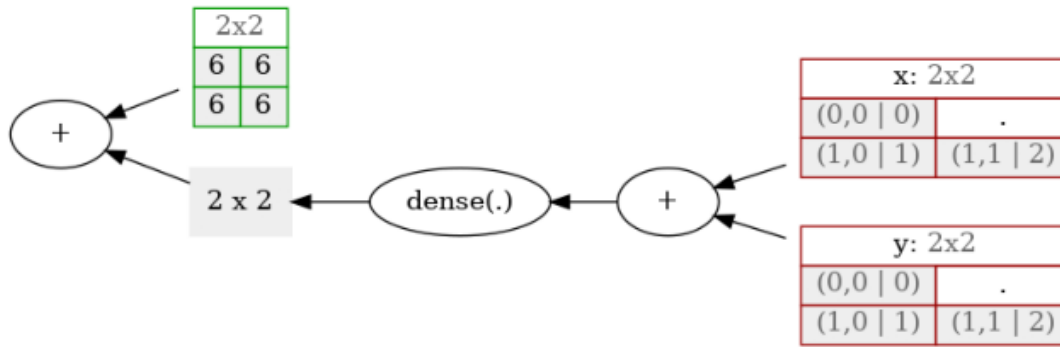
An MX graph

```
[8]: x = ca.MX.sym("x",ca.Sparsity.lower(2))
     y = ca.MX.sym("y",ca.Sparsity.lower(2))
```

```
[9]: z = ca.MX.sym("z",4,2)
```

```
[10]: zz = x+y+6
```

```
[11]: dotdraw(zz)
```



```
[12]: f = ca.Function("magic", [z,y],[z+x[0,0],x-y],{"allow_free":True})
```

```
[13]: z,z2 = f(ca.vertcat(x,y),zz.T)
```

```
[14]: z = z[:2,:] +x + ca.cos(x) - ca.sin(x) / ca.tan(z2)
```

```
[15]: dotdraw(z)
```

